Average Payoff, Typical Ticket

Unit 2 · Topic 2.8 · TODAY'S PROBLEM

Suppose a club sells 200 raffle tickets for \$2 each. One ticket wins \$100, four tickets win \$25 each, 20 tickets win \$5 each, and the other tickets win nothing. Let X be the buyer's net profit from one randomly chosen ticket: prize minus ticket cost.

Priya: *The average gross payout is \$1.50 per ticket, so buying one is a good deal.*

Mateo: *Most tickets win nothing.*

Ticket net-profit distribution

Prize	Net x	$P(X = x)$
Top prize	98	1/200
Second prize	23	4/200
Small prize	3	20/200
No prize	-2	175/200

FIRST TAKE (5 min, on your sheet)

Would you buy one ticket if your goal were to make money? Use one number or outcome from the problem.

DOK 1 **(a)** State the four possible values of X and identify the most likely value.

DOK 2 **(b)** Verify that the probabilities form a distribution. Calculate the expected gross payout and the expected net profit μ_X.

DOK 3 **(c)** Decide whether Priya's numerical statement and Mateo's statement can both be true, and explain why there is no contradiction. Assess Priya's good-deal conclusion using the ticket cost and μ_X. Explain what expected value does and does not predict for one buyer, then decide whether you would buy and defend the criterion behind your choice.

Video + follow-along: `u4_lesson7-8_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.